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Exhaust handling systems for marine vessels and related methods

Abstract

An embodiment of an exhaust handling system for a marine vessel includes a cap connected to a top end portion of an exhaust stack of the marine vessel to form an enclosure at least partially surrounding an outlet of an exhaust pipe extending through the exhaust stack. In addition, the exhaust handling system includes a collection pipe in fluid communication with the cap such that the collection pipe is to receive exhaust from the enclosure, and a coupling connected to the collection pipe that is to connect to an exhaust cleaning assembly. The exhaust cleaning system includes a tank to receive the exhaust. The cap at least partially defines a first flow path for the exhaust that extends from the enclosure to the atmosphere. The collection pipe at least partially defines a second flow path for the exhaust that extends from the enclosure to the coupling via the collection pipe.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS (1) This application is a continuation of U.S. Non-Provisional application Ser. No. 18/369,485, filed Sep. 18, 2023, titled “Exhaust Handling Systems for Marine Vessels and Related Methods,” which claims priority to and the benefit of U.S. Provisional Application No. 63/485,886, filed Feb. 18, 2023, titled “Exhaust Handling Systems for Marine Vessels and Related Methods,” and U.S. Provisional Application Ser. No. 63/488,574, filed Mar. 6, 2023, titled “Exhaust Handling Systems for Marine Vessels and Related Methods,” the contents of which are incorporated herein by reference in their entirety.

BACKGROUND

(1) This disclosure generally relates to marine vessels that travel on navigable bodies of water. More particularly, this disclosure relates to exhaust handling systems for marine vessels and related methods.

(2) A marine vessel may include any suitable vessel or boat that is transportable or movable across a navigable body of water (such as an ocean, lake, river, etc.). Such marine vessels may include engines, motors, generators, and other systems configured to output exhaust fluids (or more simply “exhaust”). Typically, the exhaust is emitted to the atmosphere during operations. However, when multiple marine vessels are concentrated in the same geographical area, the combined exhaust from the marine vessels may significantly degrade the local air quality. Berthing locations such as ports, piers, harbors, moorings, etc., may represent locations where large numbers of marine vessels congregate. Thus, in a number of jurisdictions, rules or regulations regarding the output of exhaust at berthing locations have been or will be implemented to preserve the air quality both within the berthing location and in neighboring areas.

BRIEF SUMMARY

(3) Some embodiments disclosed herein include exhaust handling systems for a marine vessel that are configured to allow selective collection of exhaust output from the exhaust-emitting systems of the marine vessel while at a berthing location. In some embodiments, the collected exhaust may be routed to an exhaust cleaning assembly that may store and/or process the exhaust to prevent harmful chemicals or pollutants within the exhaust from being emitted to the atmosphere. In some embodiments, the exhaust handling systems of the embodiments disclosed herein may include a cap that is connected to an upper end portion of an exhaust stack of the marine vessel so as to form an enclosure around one or more (such as a plurality of) exhaust pipes. The collected exhaust may be emitted directly to the atmosphere when the marine vessel is not at a berthing location (or during a situation that requires venting of exhaust to the atmosphere) or may be selectively routed to an exhaust cleaning assembly to avoid such atmospheric venting when the marine vessel is berthed. Accordingly, through use of the embodiments disclosed herein, the exhaust-emitting systems of a marine vessel may continue to operate when the marine vessel is berthed while preventing (or restricting) the emission of exhaust (or at least the harmful and/or polluting components thereof) to the atmosphere.

(4) Some embodiments disclosed herein are directed to exhaust handling system for a marine vessel. In some embodiments, the exhaust handling system includes a cap connected to a top end

portion of an exhaust stack of the marine vessel so as to form an enclosure that at least partially surrounds an outlet of an exhaust pipe extending through the exhaust stack. In addition, the exhaust handling system includes a collection pipe in fluid communication with the cap such that the collection pipe is configured to receive exhaust from the enclosure. Further, the exhaust handling system includes a coupling connected to the collection pipe that is configured to connect to an exhaust cleaning assembly. The exhaust cleaning system includes at least one tank to receive the exhaust. The cap at least partially defines a first flow path for the exhaust to flow from the enclosure to an atmosphere surrounding the cap. The collection pipe at least partially defines a second flow path for the exhaust to flow from the enclosure to the coupling via the collection pipe.

(5) In some embodiments, the exhaust handling system includes a cap connected to a top end portion of an exhaust stack of the marine vessel so as to form an enclosure that at least partially surrounds an outlet of an exhaust pipe extending through the exhaust stack. In addition, the exhaust handling system includes a vent in fluid communication with the enclosure, the vent including at least one valve member that is actuatable between a first position to emit exhaust from the enclosure to an atmosphere surrounding the cap via the vent and a second position to prevent an emission of exhaust from the enclosure to the atmosphere via the vent. Further, the exhaust handling system includes a collection pipe in fluid communication with the enclosure, and a coupling connected to the collection pipe and configured to connect to an exhaust cleaning assembly that includes at least one tank to receive the exhaust.

(6) Some embodiments disclosed herein are directed to exhaust handling system for a marine vessel. In some embodiments, the exhaust handling system includes a cap connected to a top end portion of an exhaust stack of the marine vessel such that the cap is supported by the exhaust stack and such that the cap forms an enclosure that at least partially surrounds an outlet of an exhaust pipes extending through the exhaust stack. In addition, the exhaust handling system includes a pressure-actuated vent in fluid communication with the enclosure. Further, the exhaust handling system includes a collection pipe in fluid communication with the enclosure and configured to connect to an exhaust cleaning assembly that includes at least one tank to receive the exhaust.

(7) Some embodiments are directed to methods. In some embodiment, the method includes (a) positioning a marine vessel in a berthing location, the marine vessel including a deck, an exhaust stack having an top end portion positioned above from the deck, an exhaust pipe extending through the exhaust stack to an outlet that is positioned above the top end portion of the exhaust stack, a cap connected to the exhaust stack so as to form an enclosure that at least partially surrounds the outlet of the exhaust pipe, and an collection pipe in fluid communication with the enclosure. In addition, the method includes (b) connecting an exhaust cleaning assembly to the collection pipe after (a), the exhaust cleaning assembly including at least one tank to receive the exhaust. Further, the method includes (c) operating the marine vessel to flow an exhaust emitted from the outlet of the exhaust pipe to the exhaust cleaning assembly after (b) via the collection pipe.

(8) In some embodiments, the method includes (a) positioning a marine vessel in a berthing location, the marine vessel including an exhaust stack, an exhaust pipe extending through the exhaust stack to an outlet that is positioned above a top end portion of the exhaust stack, a cap that is connected to the exhaust stack to form an enclosure that surrounds the outlet of the exhaust pipe. In addition, the method includes (b) flowing an exhaust from the enclosure to an atmosphere surrounding the cap via a first flow path during (a), the first flow path at least partially defined by the cap. Further, the method includes (c) connecting an exhaust cleaning assembly positioned at the berthing location to the cap after (a), the exhaust cleaning system including at least one tank to receive the exhaust. Still further, the method includes (d) flowing the exhaust from the enclosure to the exhaust cleaning assembly via a second flow path after (b), the second flow path at least partially defined by the cap.

(9) Embodiments described herein include a combination of features and characteristics intended to address various shortcomings associated with certain prior devices, systems, and methods. The

foregoing has outlined rather broadly the features and technical characteristics of some of the disclosed embodiments in order that the detailed description that follows may be better understood. The various characteristics and features described above, as well as others, will be readily apparent to those having ordinary skill in the art upon reading the following detailed description, and by referring to the accompanying drawings. It should be appreciated that this disclosure may be readily utilized as a basis for modifying or designing other structures for carrying out the same purposes as the disclosed embodiments. It should also be realized that such equivalent constructions do not depart from the spirit and scope of the principles disclosed herein.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

- (1) For a detailed description of various embodiments, reference will now be made to the accompanying drawings in which:
- (2) FIG. 1 is a schematic side view of a marine vessel having an exhaust handling system according to some embodiments of the disclosure;
- (3) FIG. 2 is a schematic rear view of the marine vessel of FIG. 1 according to some embodiments of the disclosure;
- (4) FIG. 3 is a perspective view of a cap of the exhaust handling system of the marine vessel of FIG. 1 according to some embodiments of the disclosure;
- (5) FIG. 4 is a cross-sectional view taken along section A-A in FIG. 3 according to some embodiments of the disclosure;
- (6) FIGS. 5 and 6 are side, cross-sectional views of a vent for a cap of the exhaust handling system of FIG. 3 according to some embodiments of the disclosure;
- (7) FIG. 7 is a schematic view of the marine vessel of FIG. 1 connected to an exhaust cleaning assembly positioned on a barge according to some embodiments of the disclosure;
- (8) FIG. 8 is a schematic view of the marine vessel of FIG. 1 connected to an exhaust cleaning assembly positioned on-shore according to some embodiments of the disclosure;
- (9) FIG. 9 is a cross-sectional view taken along section A-A in FIG. 3 according to some embodiments of the disclosure;
- (10) FIG. 10 is a schematic view of the cap of the exhaust handling system of the marine vessel of FIG. 1 according to some embodiments of the disclosure;
- (11) FIGS. 11A-11D are sequential top views of a sequence or method for routing exhaust from the marine vessel of FIG. 1 via the exhaust handling system when the marine vessel is berthed at a berthing location and when the marine vessel is traveling into and out of the berthing location according to some embodiments of the disclosure;
- (12) FIG. 12 is a schematic view of the exhaust handling system of the marine vessel of FIG. 1 further showing a controller for controlling the flow of exhaust from the marine vessel via the exhaust handling system according to some embodiments of the disclosure; and
- (13) FIGS. 13 and 14 are diagrams of methods for flowing the exhaust of a marine vessel through an exhaust handling system according to some embodiments of the disclosure.

DETAILED DESCRIPTION

- (14) As previously described, the exhaust from marine vessels at a berthing location may degrade air quality and may even be restricted by local rules and regulations. However, ceasing operation of all exhaust-emitting systems on a marine vessel may not be desirable or feasible while the marine vessel is at the berthing location. For instance, electrical generation systems (for example, diesel generators, turbine generators, etc.) may continue to operate so that other electrically operated systems and assemblies of the marine vessel (for example, communications systems, safety systems, control systems, water and sewage systems, HVAC systems, etc.) may also be operated

while at the berthing location. In addition, some berthing locations do not have sufficient infrastructure to support and operate these various systems and assemblies of the berthed marine vessels independently of the onboard exhaust-emitting systems.

(15) Accordingly, embodiments disclosed herein include exhaust handling systems for a marine vessel that are configured to allow selective collection of exhaust output from the exhaust-emitting systems of the marine vessel while at a berthing location. In some embodiments, the collected exhaust may be routed to an exhaust cleaning assembly that may store and/or process the exhaust to prevent harmful chemicals or pollutants within the exhaust from being emitted to the atmosphere. In some embodiments, the exhaust handling systems of the embodiments disclosed herein may include a cap that is fixed to an upper end of an exhaust stack of the marine vessel so as to form an enclosure around one or more (e.g., a plurality of) exhaust pipes. The collected exhaust may be emitted directly to the atmosphere when the marine vessel is not at a berthing location (or during a situation that requires venting of exhaust to the atmosphere) or may be selectively routed to an exhaust cleaning assembly to avoid such atmospheric venting when the marine vessel is berthed. Accordingly, through use of the embodiments disclosed herein, the exhaust-emitting systems of a marine vessel may continue to operate when the marine vessel is berthed while preventing (or restricting) the emission of exhaust (or at least the harmful and/or polluting components thereof) to the atmosphere.

(16) Reference is now made to FIG. 1, which shows a marine vessel **10** and an exhaust handling system **100** according to some embodiments. The marine vessel **10** may include any suitable vessel or ship that may travel within or across a navigable body of water such as, for instance, an ocean, a sea, a lake, a river, a channel, etc. In some embodiments, the marine vessel **10** may be a cargo vessel, such as a container ship, tankship, reefer ship, etc.

(17) Marine vessel **10** includes a rear end (or stern) **10a**, a front end (or bow) **10b** opposite rear end **10a**, and a main deck (or more simply “deck”) **12** extending between ends **10a**, **10b**. Deck **12** may define an exterior surface (or collection of exterior surfaces) on the marine vessel **10** that may be accessed by personnel.

(18) A rudder **14** and propeller **16** may be positioned at (or proximate to) rear end **10a**. As is known to one having ordinary skill in the art, the propeller **16** may provide propulsion to the marine vessel **10**, and the rudder **14** may be turned to steer the marine vessel **10** within a body of water.

(19) An accommodation deck (or more simply “accommodation”) **18** is positioned atop deck **12**. The accommodation **18** is the living space of the marine vessel **10** and may include one or more cabins (or rooms), galleys, store rooms, messrooms, or other rooms that may be used by personnel. A bridge **20** may be positioned atop (or adjacent to) the accommodation **18**. The bridge **20** may include one or more controls for the marine vessel **10** including (for instance) steering controls, communications systems, other system controls, etc.

(20) As shown in FIGS. 1 and 2, an exhaust stack **50** may extend upward and away from deck **12**. The exhaust stack **50** may sometimes be referred to as an “exhaust funnel.” The exhaust stack **50** may be positioned adjacent the accommodation **18** and bridge **20** along deck **12**. Specifically, the exhaust stack **50** may be positioned between bridge **20** and rear end **10a** along deck **12** in some embodiments. Exhaust stack **50** may extend upward from deck **12** to a top end portion **50a**. One or more (such as one or a plurality of) exhaust pipes **52** extend through exhaust stack **50** and out of the top end portion **50a**. Specifically, each exhaust pipe **52** has an outlet **54** that is extended out of the exhaust stack **50** and positioned above the top end portion **50a**.

(21) During operations, each exhaust pipe **52** (or some of the exhaust pipes **52**) may output exhaust fluid (or “exhaust”). Specifically, as shown in FIG. 2, each exhaust pipe **52** is fluidly connected to one or more exhaust-emitting systems **62**, **64**, **66** (collectively referred to as “exhaust-emitting systems **60**”). For instance, the exhaust-emitting systems **60** of marine vessel **10** may include diesel electric generator(s) **62**, boilers **64**, inert gas systems **66**, or other systems or assemblies that may output combustion (or other) exhaust during operations. Accordingly, marine vessel **10** includes an

exhaust handling system **100** that is configured to selectively collect and route exhaust emitted from the exhaust pipes **52** to an exhaust cleaning assembly (not shown in FIG. **1** but see exhaust cleaning assembly **200** shown in FIGS. **7** and **8** and described herein).

(22) As shown in FIGS. **1** and **2**, exhaust cleaning assembly **100** includes a cap **110** that is connected to exhaust stack **50**. Specifically, cap **110** may be connected to top end portion **50a** of exhaust stack **50** so as to form or define an enclosure **112** that surrounds (or at least partially surrounds) the outlets **54** of each of the exhaust pipes **52**. As will be described in more detail below, the enclosure **112** may collect exhaust emitted from the outlets **54** of exhaust pipes **52**. A collection pipe **150** extends from cap **110** toward the deck **12** and is connected to the cap **110** such that the collection pipe **150** is in fluid communication with the enclosure **112**. Thus, during operations, exhaust may be collected within enclosure **112** and routed toward deck **12** via collection pipe **150**.

(23) In some embodiments, the cap **110** may be connected to the exhaust stack **50** so that the outlets **54** of less than all of the exhaust pipes **52** are surrounded (at least partially) by the enclosure **112**. Thus, in some embodiments, the outlets **54** of one or more of the exhaust pipes **52** may be positioned outside of the enclosure **112** and the outlets of one or more of the exhaust pipes **52** may be positioned inside the enclosure **112**. For instance, without being limited to this or any other theory, outlet pipes **52** associated with the main engine of the marine vessel **10** may not normally emit exhaust (or may not emit a substantial volume of exhaust) when the marine vessel **10** is stationary at a berthing location. Thus, the outlet pipes **52** associated with the main engine of the marine vessel **10** may not be positioned in the enclosure **112** so as to minimize a size and complexity of the cap **110** as well as to avoid constriction of the exhaust flow out from the main engine of the marine vessel **10** when the marine vessel **10** is not at a berthing location. In some embodiments, multiple caps **110** may be attached to exhaust stack **50** (or to multiple exhaust stacks **50** depending on the configuration of the marine vessel **10**) so that different caps **110** may surround (at least partially) different ones or groups of the exhaust pipes **52** during operations. For instance, in some embodiments, a first cap **110** may be positioned on a first exhaust stack **50** so as to surround (at least partially) one or more exhaust pipes **52** extending therethrough, and a second cap **110** may be positioned on a second exhaust stack **50** so as to surround (at least partially) one or more exhaust pipes **52** extending therethrough.

(24) Collection pipe **150** (or a portion of collection pipe **150**) may include a rigid pipe that is rigidly connected (such as via brackets, welding, bolting, riveting, etc.) to cap **110** and marine vessel **10** (FIGS. **1** and **2**). In some embodiments, the collection pipe **150** (or a portion of the collection pipe **150**) may include flexible ductwork (or ducting) such as, for instance, flexible hoses and the like that may be readily deformed and maneuvered during operations. In some embodiments, the collection pipe **150** may be temporarily connected to the cap **110** and/or the marine vessel **10** so as to allow collection pipe **150** to be easily removed when not in use (such as when marine vessel **10** is not at a berthing location). In addition, in some embodiments, the collection pipe **150** (whether it includes rigid pipe and/or flexible ductwork) may include one or more layers of thermal insulation.

(25) Collection pipe **150** may include or be connected to a coupling **152** that, as will be described in more detail below, may be connected to an exhaust cleaning assembly during operations. The coupling **152** may be positioned at or proximate to the deck **12** such that the coupling **152** may be accessible from the deck **12**. Thus, during operations, personnel may interact with the coupling **152** (for example, to connect the coupling **152** and collection pipe **150** to an exhaust cleaning assembly) from the deck **12**. Further details of the exhaust handling system **100** are now described below according to some embodiments.

(26) As shown in FIGS. **3** and **4**, in some embodiments the cap **110** may include an upper closed end **110a** (or more simply “upper end **110a**”) and a lower open end **110b** (or more simply “lower end **110b**”) opposite the upper end **110a**. In some embodiments, the cap **110** may be shaped as a rectangular parallelepiped, such that the cap **110** includes a planar top **116** and a plurality of (e.g.,

four) planar sides **114** extending from planar top **116** to lower end **110b**. However, other shapes are contemplated for cap **110**, such as, for instance, cylindrical, conical, triangular prism, irregular shape, etc. As best shown in FIG. 4, the top **116** and sides **114** define the enclosure **112**. During operations, cap **110** may be connected to top end portion **50a** of exhaust stack **50** such that the enclosure **112** surrounds the outlets **54** of the exhaust pipes **52** as previously described. In addition, the collection pipe **150** may be fluidly connected to the enclosure **112** on one of the planar sides **114**.

(27) A plurality of flow paths are defined within exhaust handling system **100** to route exhaust out of the enclosure **112** of cap **110** during operation. For instance, as shown in FIG. 4, one or more first flow paths **122** may allow exhaust to flow from the enclosure **112** to the atmosphere **123** surrounding the cap **110**. For instance, the first flow path(s) **122** may extend out of the enclosure **112** and through one or more vents **120**, **160** to the surrounding atmosphere **123**. Thus, the first flow path(s) **122** may be at least partially defined by the cap **110** and one or more of the vents **120**, **160**.

(28) The vents **120** are connected to the cap **110** such that they are in fluid communication with the enclosure **112**. The vents **120** may include an actuatable vent. For example, in some embodiments, the vents **120** may have a valve or valve member **125** (such as a gate valve, flapper valve, butterfly valve, etc.) that is actuatable between an open position and a closed position. When the valve member **125** of a vent **120** is in the open position, the vent **120** may allow exhaust to flow therethrough and into the surrounding atmosphere **123**, and when the valve member **125** of a vent **120** is in the closed position, the vent **120** may prevent (or restrict) the flow of exhaust therethrough to the surrounding atmosphere **123**. In some embodiments, the valve members **125** of vents **120** may be actuated between the open and closed positions by a controller (such as controller **300** shown in FIG. 12 and described herein) and/or manually by personnel. In some embodiments, the vents **120** (more specifically the valve members **125**) may be pressure actuated (such that the vents **120** may be “pressure-actuated vents”). Specifically, in some embodiments, the valve member **125** of one or more of the vents **120** may be biased such that the valve member **125** is configured to open when a sufficient differential pressure is applied across the vent **120** (and against the bias applied to the valve member **125**). For instance, in some embodiments, the valve member(s) **125** of one or more of the vents **120** may be biased to the closed position (for example, via a spring or other suitable biasing member or assembly), and when a pressure within the enclosure **112** rises above a threshold, the valve member(s) **125** of the one or more of the vents **120** may transition from the closed position to the open position to allow exhaust to flow out of the enclosure **112** to the surrounding atmosphere **123**.

(29) The vent **160** may be connected to cap **110** such that vent **160** is in fluid communication with the enclosure. The vent **160** may be positioned along the planar top **116** of cap **110** and may be also be configured to transition between a closed position and an open position to selectively prevent and allow exhaust to flow out of enclosure **112** to the surrounding atmosphere **123**, respectively. As shown in FIG. 5, in some embodiments, the vent **160** may include a housing **162** that defines an opening or flow path **164** therethrough. The housing **162** is mounted to the planar top **116** so that the opening **164** forms or defines an opening in the planar top **116** of cap **110**. A plurality of louvers **166** are pivotably connected to the housing **162** such that the louvers **166** are parallel to one another and span across the opening **164**. Each louver **166** is pivotably connected to the housing **162** via a corresponding hinge **168** so that the louvers **166** may pivot about the hinge **168**, relative to the housing **162** during operations. Specifically, the louvers **166** may be pivoted about the hinges **168** between a first or closed position shown in FIG. 5 and a second or open position shown in FIG. 6 (as well as a plurality of positions between the closed position of FIG. 5 and the open position of FIG. 6). Thus, the vent **160** may be referred to herein as an “actuatable vent” similar to embodiments of the vent **120** previously described, and the louvers **166** may be referred to as actuatable “valve members” or “valves” similar to embodiments of the valve members **125** of vents

120 previously described.

(30) When the louvers **166** are in the closed position (FIG. 5), the louvers **166** may engage, interlock, overlay, overlap, and/or otherwise cooperate with one another so as to cover and therefore occlude the opening **164**. As a result, when the louvers **166** are in the closed position (FIG. 5), exhaust is prevented (or is at least restricted) from flowing out of the enclosure **112** to the surrounding atmosphere **123**. In some embodiments, the louvers **166** may include or be connected to seals (such as compliant seals) that may further enhance the ability of louvers **166** to prevent (or at least restrict) the flow of exhaust out of the enclosure **112** via the opening **164** when louvers **166** are in the closed position (FIG. 5). Conversely, when the louvers **166** are in the opening position (FIG. 6), the louvers **166** may rotate or pivot about the hinges **168** so as to disengage with one another and thereby open or uncover the opening **164** of housing **162** and place the surrounding atmosphere **123** in fluid communication with the enclosure **112**.

(31) The louvers **166** may be synchronously rotated about the hinges **168** between the closed position (FIG. 5) and the open position (FIG. 6) by a suitable transmission or connection **163** (such as a connection bar, gears-such as rack and pinion gears, or other suitable connection device or assembly). A driver **170** may be connected to the louvers **166** (such as directly connected to louvers **166** or indirectly connected to louvers **166** via connection **163**). In some embodiments, the driver **170** may include a suitable motor, such as, for instance, an electric motor, hydraulic motor, pneumatic motor, etc. In some embodiments, the driver **170** may include a manual driver such as a chain pull, lever, or other manually operated device. During operations, the driver **170** may actuate the louvers **166** (such as via the connection **163**) to rotate or transition between the closed position (FIG. 5) and the closed position (FIG. 6) to selectively prevent (or restrict) or allow, respectively, exhaust to flow out of the enclosure **112** to the surrounding atmosphere **123**. In some embodiments, the vent **160** may also include a fan or blower that may draw exhaust out of the enclosure **112** and through the opening **164** and into the atmosphere **123** when the louvers **166** are in the open position (FIG. 6). Still further, in some embodiments, the louvers **166** may be actuated between the open position and the closed position by a differential pressure applied across the vent **160** between the enclosure **112** and the atmosphere **123**. For instance, the driver **170** may be connected to or may include a pressure sensor and may actuate the louvers **166** based on an output thereof. As another example, the louvers **166** may be rotationally biased (for example, via torsional springs or other biasing members or systems) toward the closed position (FIG. 5) and may transition to the open position (FIG. 6) in response to a sufficient differential pressure between the enclosure **112** and atmosphere **123**. Thus, the vent **160** may also be referred to herein as a “pressure-actuated vent” as described herein.

(32) As shown in FIG. 4, in some embodiments, first flow path(s) **122** may extend through one or more of the vents **120**, **160** positioned on the planar top **116** of the cap **110**. In some embodiments, the first flow path(s) **122** may extend through vent(s) **120** and/or vents **160** that are positioned along collection pipe **150**. In some embodiments, a plurality of first flow paths **122** may extend out of the enclosure **112** via vents **120**, **160** in the planar top **116** and/or the collection pipe **150**; however, in some embodiments, a single first flow path **122** may extend out of the enclosure **112** via a vent **120** or a vent **160** in either the planar top **116** or the collection pipe **150**. Still further, in some embodiments, a first flow path **122** may extend out of the enclosure **112** via a vent **120** and/or a vent **160** that is positioned along and/or coupled to one of the planar sides **114** of cap **110**.

(33) As shown in FIG. 4, one or more second flow paths **124** may allow exhaust to flow out of the enclosure **112** via the collection pipe **150**. Thus, the one or more second flow paths **124** may be at least partially defined by the cap **110** and the collection pipe **150**. As previously described, the collection pipe **150** may extend from the cap **110** to the coupling **152** (that may be accessible from the deck **12** of marine vessel **10** as previously described). In particular, the coupling **152** may include any suitable coupling mechanism that is configured to connect the collection pipe **150** to an exhaust cleaning assembly (such as the exhaust cleaning assembly **200** shown in FIGS. 7 and 8) via

a suitable conduit (for example, a hose, pipe, tubing, etc.). For instance, in some embodiments, the coupling **152** may include a flanged coupling, quick connect coupling, threaded coupling, union, clamped coupling, or some combination thereof. A valve **154** is positioned along the collection pipe **150** between the enclosure **112** and the coupling **152**. The valve **154** may be actuated between an open position to allow exhaust to flow along collection pipe **150** via second flow path **124** to coupling **152**, and a closed position to prevent (or at least restrict) the flow of exhaust along the collection pipe **150** via second flow path **124** to coupling **152**. In some embodiments, the valve **154** may be actuated between the open and closed positions by a controller (such as the controller **300** shown in FIG. **12**) and/or manually by personnel.

(34) A cleanout port **156** may be positioned along the collection pipe **150** between the coupling **152** and the enclosure **112**. Specifically, the cleanout port **156** may be positioned along collection pipe **150** so that it is accessible from the deck **12** of marine vessel **10**. The cleanout port **156** is configured to provide access into the collection pipe **150** independent of the coupling **152** so that personnel may clean out or remove debris that may collect within the collection pipe **150** during operations and thereby prevent the second flow path **124** (or one or more of the first flow paths **122**) from becoming obstructed. In some embodiments, the clean out port **156** may be closed or occluded via a flanged cap **157**; however, any suitable capping or closing device (such as a hatch, blind, etc.) may be utilized to close the cleanout port **156** in some embodiments.

(35) In some embodiments, the cap **110** may be permanently or fixedly installed on the marine vessel **10**. Specifically, the cap **110** may remain connected and fixed to the exhaust stack **50** both when the marine vessel is at berth (such as at a berthing location) and when the marine vessel **10** is not at a berthing location and is moving across a body of water (for example, across an ocean or lake). Thus, the cap **110** may remain fixed to the exhaust stack **50** so that the cap **110** may not be readily lifted or removed from the exhaust **50** without breaking or disconnecting the connections between the cap **110** and exhaust stack **50**. In some embodiments, the cap **110** may be welded to the exhaust stack **50**. In some embodiments, the cap **110** may be integrally formed as part of the exhaust stack **50** itself. In some embodiments, the cap **110** may be fixed to the exhaust stack **50** via one or more of a bolted connection, a riveted connection, or other suitable connections or structures.

(36) In some embodiments, the cap **110** may be temporarily connected to the exhaust stack **50** such that the cap **110** may be installed on the exhaust stack **50** when the marine vessel **10** is positioned at a berthing location and then may be removed when the marine vessel **10** is to move out of (or away from) the berthing location. In some embodiments, the cap **110** may be temporarily installed on the exhaust stack **50** via clamps, a shouldered engagement (such as by resting the cap **110** on an external shelf or shoulder of the exhaust stack **50**), and/or any other suitable temporary connection.

(37) Regardless as to whether the cap **110** is permanently or temporarily connected to the exhaust stack **50**, in some embodiments, the cap **110** may be supported (such as fully supported) by the exhaust stack **50**. That is, the weight of the cap **110** may be borne by the exhaust stack **50** (or other frames, structures, or other components of the marine vessel **10**) during operations so that a crane or other lifting device may not be attached to the cap **110** during operations. A crane (or other lifting device) may be utilized to lower the cap **110** onto exhaust stack **50** or lift the cap **110** away from the exhaust stack **50**; however, once the cap **110** is connected to the exhaust stack **50**, the crane (or other lifting device) may be disconnected from cap **110** and the weight of the cap **110** may be borne by the exhaust stack **50** and/or other portions or components of the marine vessel **10**.

(38) The flow paths **122**, **124** may selectively route exhaust out of the enclosure **112** when the marine vessel **10** is and is not at berth. For instance, during operations, when the marine vessel **10** is not at berth (such as when marine vessel **10** is moving across or within a body of water), exhaust emitted from the outlets **54** of exhaust pipes **52** may flow into the enclosure **112** and then is emitted from the enclosure **112** via the one or more first flow paths **122**. Specifically, when the marine vessel **10** is not at berth, the valve **154** positioned along the collection pipe **150** may be transitioned

to the closed position to prevent exhaust from flowing out of enclosure **112** and toward the deck **12** via the coupling **152**. As a result, the exhaust emitted from the outlets **54** of exhaust pipes **52** may flow out of the enclosure **112** to the surrounding atmosphere **123** via the vent(s) **120** and/or the vent(s) **160** along the first flow path(s) **122**. As previously described, the vents **120**, **160** may be actuated (e.g., via controller, personnel, etc.) to the open position to allow the exhaust to flow out of the enclosure **112** to the surrounding atmosphere **123**. Alternatively, for embodiments in which the vents **120**, **160** are pressure-actuated as previously described, the emission of the exhaust from the outlets **54** of exhaust pipes **52** into enclosure **112** increases the pressure within the enclosure **112** (e.g., such as when the valve **154** is closed) until one or more of the vent(s) **120**, **160** are transitioned to the open position to vent the exhaust to the surrounding atmosphere **123** along the first flow path(s) **122**.

(39) As shown in FIGS. **4** and **7**, when the marine vessel **10** is at berth (for example, is moored or docked at a berthing location as previously described), the exhaust emitted into the enclosure **112** via the outlets **54** of exhaust pipes **52** may be flowed out of the enclosure **112** via the second flow path **124**. Specifically, when the marine vessel **10** is at berth, an exhaust cleaning assembly **200** may be connected to the coupling **152** of the collection pipe **150** via a conduit **202**, and valve **154** (FIG. **4**) may be transitioned to the open position. As a result, exhaust emitted from outlets **54** of exhaust pipes **52** may be flowed out of the enclosure **112** to the exhaust cleaning assembly **200** via the second flow path **124** and conduit **202**. As is known by one having ordinary skill in the art, within the exhaust cleaning assembly **200**, the exhaust may be treated via one or more suitable processes or assemblies to remove some or all of the pollutants or other harmful constituents of the exhaust (e.g., nitrous oxide (NO_x), carbon dioxide (CO₂), carbon monoxide (CO), soot, etc.). In some embodiments, the exhaust cleaning assembly **200** may simply capture and store the exhaust such that the exhaust may then be transferred to a suitable cleaning process or other storage facility. Thus, in some embodiments, the exhaust cleaning assembly **200** may include one or more tanks (or other suitable vessels **204** that are configured to receive and store the exhaust (or some other fluid such as treated exhaust, treatment fluids, etc.). Additionally or alternatively, in some embodiments, the exhaust cleaning system **200** may include one or more reactors **206** such as catalyst reactors, that are configured to treat or clean the exhaust during operations.

(40) Regardless of the particular makeup or function of the exhaust cleaning assembly **200** (such whether the exhaust cleaning assembly **200** is configured to treat or simply store exhaust during operations), the exhaust emitted from the outlets **54** of exhaust pipes **52** may be prevented from flowing (or mostly restricted from flowing) to the atmosphere **123** at the berthing location when coupling **152** is connected to exhaust cleaning assembly **200** via conduit **202**. For instance, when the exhaust is flowing along the second flow path **124** to the exhaust cleaning assembly, the vent(s) **120**, **160** may be actuated to the closed position (such as via a controller or personnel or by a differential pressure between the atmosphere **123** and enclosure **112**/collection pipe **150**) to prevent a flow of exhaust along the first flow path(s) **122**. As a result, the atmospheric conditions at the berthing location may be improved without having to shut down the exhaust-emitting systems **60** (FIG. **2**) of the marine vessel **10**.

(41) As shown in FIG. **7**, in some embodiments, the exhaust cleaning assembly **200** may be positioned on at barge or other marine vessel that is positioned proximate to (e.g., alongside) the marine vessel **10** while the marine vessel **10** is at the berthing location. As shown in FIG. **8**, in some embodiments, the exhaust cleaning assembly **200** may be positioned onshore at the berthing location. In some embodiments, the exhaust cleaning assembly **200** (or a portion thereof) may be remote from the berthing location (e.g., one or more miles away from the berthing location) and the conduit **202** may include (or be connected to) a suitable pipeline or other suitable infrastructure to transport the exhaust from the berthing location to the remotely located exhaust cleaning assembly **200**.

(42) As shown in FIG. **9**, as previously described, in some embodiments, exhaust is output from the

outlets **54** of the exhaust pipes **52** into the enclosure **112**, and then the exhaust may be flowed out of the enclosure **112** via the one or more first flow paths **122** or one or more second flow paths **124**. However in some embodiments, the collection pipe **150** may be individually connected to the outlets **54** of one or more of the exhaust pipes **52** via a plurality of pipe connections **158**. The pipe connections **158** (or more simply “connections” **158**) may include any suitable conduit and/or connector that may connected to and thus in fluid communication with the outlets **54** to the collection pipe **150**. For instance, in some embodiments, the pipe connections **158** may include pipes, hoses, tubing, or some combination thereof. During operations with the embodiment depicted in FIG. **9**, exhaust emitted from the outlets **54** of the exhaust pipes **52** may flow directly into the collection pipe **150** via the plurality of pipe connections **158** and may thus bypass (and not flow into) the enclosure **112** defined by cap **110**. In some embodiments, the pipe connection **158** may route exhaust from one or more of the exhaust pipes **52** separately from exhaust from one or more others of the exhaust pipes **52**. For instance, one or more of the exhaust pipes **52** may output exhaust to an exhaust cleaning system **200** via the pipe connections **158** and collection pipe **150** and one or more others of the exhaust pipes **52** may output exhaust to the atmosphere **123** and/or another exhaust cleaning system **200** via the enclosure **112** and another collection pipe **150** (not shown) connected to cap **110**.

(43) As is shown in FIG. **9**, for embodiments that include pipe connections **158** for individually connecting the outlet **54** of exhaust pipes **52** to collection pipe **150**, the one or more first flow paths **122** may extend from the collection pipe **150** through one or more vents **120** (or one or more vents **160**) connected to collection pipe **150**. Thus, in these embodiments, the first flow path(s) **122** may bypass the enclosure **112**. Vent(s) **120**, **160** may also be positioned on the cap **110** (e.g., along planar top **116**) to allow exhaust (or other fluid) to vent from enclosure **112** (e.g., in the event the exhaust is leaking from pipes **52**, connections **158**, collection pipe **150**, etc.). In addition, for the embodiment of FIG. **9**, the second flow path **124** may extend though the collection pipe **150** toward the deck **12** and coupling **152** as shown in FIG. **4** and previously described.

(44) As shown in FIG. **10**, in some embodiments, each pipe connection **158** may independently route the exhaust emitted from the corresponding exhaust pipe **52** out of the enclosure **112** of cap **110** via a separate collection pipe **159**. Thus, in these embodiments, the common collection pipe **150** may be omitted. However, in some embodiments, each of the separate collection pipes **159** (or some of the collection pipes **159**) may be nested within the common collection pipe **150**. As shown in FIG. **10**, in some embodiments, the separate collection pipes **159** may independently route exhaust from the outlets **54** of exhaust pipes **52** to the surrounding atmosphere **123** or to one or more exhaust cleaning assemblies (such as, exhaust cleaning assembly **200**). For instance, within each of the separate collection pipes **159**, a first flow path **122** may be defined that routes exhaust from the corresponding collection pipe **159** and out to the surrounding atmosphere **123** via a vent **120** that is connected to the corresponding collection pipe **159**. In addition, within each of the separate collection pipes **159**, a second flow path **124** may be defined that routes exhaust from the corresponding collection pipe **159** toward a coupling **152** connected to the corresponding collection pipe **159** and accessible from the deck **12** of the marine vessel **10** (FIG. **4**). Each of the couplings **152** (or some of the couplings **152**) connected to the separate collection pipes **159** may be connected (such as, via a conduit such as conduit **202** shown in FIGS. **4** and **7**) to an exhaust cleaning assembly (e.g., exhaust cleaning assembly **200** shown in FIGS. **4** and **5**) when the marine vessel **10** is at a berthing location as previously described.

(45) Reference is now generally made to FIGS. **11A-11D**, in which an example sequence or method for routing exhaust from the marine vessel **10** via the exhaust handling system **100** when the marine vessel **10** is positioned at a berthing location **250** (such that the marine vessel **10** is berthed) and when the marine vessel **10** is traveling into and out of the berthing location **250** is shown according to some embodiments. In describing the sequence shown in FIGS. **11A-11D**, continuing reference will be made to the various features of embodiments of marine vessel **10** and exhaust

handling system **100** shown in FIGS. **1-10** and previously described above.

(46) Initially, as shown in FIGS. **11A** and **11B**, the marine vessel **10** may travel toward the berthing location **250**. For instance, as previously described, the berthing location **250** may include a port, dock, harbor, mooring, etc., and the marine vessel **10** may travel to the berthing location **250** for any suitable reason (such as to offload or receive cargo, undergo repairs, refuel, offload or receive personnel, undergo inspection, etc.). When the marine vessel **10** is traveling (such as across a body of water such as an ocean, lake, channel, etc.), the exhaust emitted from exhaust stack **50** (particularly from the exhaust pipes **52** extending through exhaust stack **50**), may be vented to the surrounding atmosphere **123** via the first flow path(s) **122** (FIG. **4**) extending out of the cap **110**. An exhaust cleaning assembly **200** may be positioned at (or accessible from) the berthing location **250** as previously described. For instance, as previously described, the exhaust cleaning assembly **200** may be positioned on a barge that is floating on the water within the berthing location **250**, or the exhaust cleaning assembly **200** may be positioned onshore at the berthing location **250** (or at a remote location from the berthing location **250** and accessible via pipeline or other suitable infrastructure as previously described).

(47) As shown in FIG. **11C**, once the marine vessel **10** is positioned within the berthing location **250**, the exhaust handling system **100** may be fluidly connected to the exhaust cleaning assembly **200** via coupling **152** and conduit **202** as previously described. As a result, while the marine vessel **10** is berthed, any exhaust emitted from the exhaust stack **50** (particularly from the one or more exhaust pipes **52**) may be flowed to the exhaust cleaning assembly **200** via the exhaust handling system **100** to be stored (such as in the one or more tanks **204**), cleaned (such as via the one or more reactors **206**), or otherwise processed as previously described. Thereafter, as shown in FIG. **11D**, when the time comes for marine vessel **10** to depart from the berthing location **250**, the exhaust cleaning assembly **200** may be disconnected from the exhaust handling system **100**, and the marine vessel **10** may travel out of and away from the berthing location **250**. Once the exhaust handling system **100** is disconnected from exhaust cleaning assembly **200** and while marine vessel **10** is traveling out of and away from the berthing location **250**, any exhaust emitted from exhaust stack **50** (particularly the one or more exhaust pipes **52**) may once again be emitted to the surrounding atmosphere **123** via the first flow path(s) **122** (FIG. **4**) extending out of the cap **110**.

(48) Thus, exhaust emitted from the exhaust stack **50** of marine vessel **10** may be continuously vented via the exhaust handling system **100** both when the marine vessel **10** is berthed (such as at the berthing location **250**) and when the marine vessel **10** is away from the berthing location **250** (and traveling to and away therefrom). By permanently fixing and integrating the exhaust handling system **100** (including cap **110** and collection pipe **150**) onto the marine vessel **10**, personnel may simply connect the conduit **202** to the coupling **152** upon arrival at the berthing location **250**, thereby eliminating the need to use heavy-lift equipment, such as a crane, to transfer or place a temporary cap atop the exhaust stack **50** once the marine vessel **10** arrives at the berthing location **250**. Accordingly, when the marine vessel **10** is berthed (such as at a berthing location) exhaust-emitting systems (such as exhaust-emitting systems **60** shown in FIG. **2**) may continue to operate so as to support operation of the marine vessel **10** and any sub-systems thereof, and exhaust may be prevented (or restricted) from being emitted into the surrounding atmosphere **123** within the berthing location **250**. As a result, the air quality of a berthing location **250** may be maintained, even is multiple marine vessels **10** are positioned therein.

(49) As shown in FIG. **12**, in some embodiments, a controller **300** may be used to control the flow of exhaust through the exhaust handling system **100** during operation of the marine vessel **10** both when the marine vessel **10** is berthed (for example, at berthing location **250** shown in FIGS. **11A-11D**) and when the marine vessel **10** is not berthed. The controller **300** may be (or may be incorporated within) a main or master controller onboard the marine vessel **10**, or the controller **300** may be a standalone controller **300** for controlling the flow of exhaust through the exhaust handling system **100**. In either case, the controller **300** may be described and referred to herein as being a

part of the exhaust handling system **100**. The controller **300** may be positioned onboard the marine vessel **10**, such as on the bridge **20**; however, controller **300** may be positioned at any location (or distributed among multiple locations) onboard or off the marine vessel **10**.

(50) The controller **300** may be a computing device, such as a computer, tablet, smartphone, server, or other computing device or system. Thus, controller **300** may include a processor **302** and a memory **304**. The processor **302** may include any suitable processing device or a collection of processing devices. In some embodiments, the processor **302** may include a microcontroller, central processing unit (CPU), graphics processing unit (GPU), timing controller (TCON), scaler unit, or some combination thereof. During operations, the processor **302** executes machine-readable instructions (such as machine-readable instructions **306**) stored on memory **304**, thereby causing the processor **302** to perform some or all of the actions attributed herein to the controller **300**. In general, processor **302** fetches, decodes, and executes instructions (e.g., machine-readable instructions **306**). In addition, processor **302** may also perform other actions, such as, making determinations, detecting conditions or values, etc., and communicating signals. If processor **302** assists another component in performing a function, then processor **302** may be said to cause the component to perform the function.

(51) The memory **304** may be any suitable device or collection of devices for storing digital information including data and machine-readable instructions (such as machine-readable instructions **306**). For instance, the memory **305** may include volatile storage (such as random access memory (RAM)), non-volatile storage (e.g., flash storage, read-only memory (ROM), etc.), or combinations of both volatile and non-volatile storage. Data read or written by the processor **302** when executing machine-readable instructions **306** can also be stored on memory **304**. Memory **304** may include “non-transitory machine-readable medium,” where the term “non-transitory” does not include or encompass transitory propagating signals.

(52) The processor **302** may include one processing device or a plurality of processing devices that are distributed within controller **300** or more broadly within marine vessel **10**. Likewise, the memory **304** may include one memory device or a plurality of memory devices that are distributed within controller **300** or more broadly within marine vessel **10**.

(53) The controller **300** may be communicatively connected (such as via wired and/or wireless connection) to a user interface **308** (such as a monitor, display, computing device, touch-sensitive screen or other surface, keyboard, mouse, or some combination thereof). During operations, a user (e.g., personnel onboard the marine vessel **10**) may view information output from the controller **300** on the user interface **308** (such as the position or status of one or more of the sensors **320**, **322**, vent(s) **120**, vent(s) **160**, valve **154**, coupling **152**, etc.). In addition, during operations, a user may make inputs to the controller **300** via the user interface **308** (such as commands to open valve **154** and/or vent(s) **120**, **160**).

(54) Controller **300** may be connected to various sensors (such as sensors **320**, **322**) positioned throughout the exhaust handling system **100**. For instance, controller **300** may be connected to a pressure sensor **320** that is connected to the cap **110** and configured to detect or measure a pressure within the enclosure **112** (or value indicative thereof). In addition, controller **300** may be connected to a pressure sensor **322** that is connected to the collection pipe **150** and configured to detect or measure a pressure within the collection pipe **150** (or a value indicative thereof). The pressure sensors **320**, **322** may be any suitable device that is configured to measure, detect, or determine a pressure (or value indicative thereof) within a given area, volume, location. For instance, in some embodiments, the pressure sensors **320**, **322** may include strain gauges, capacitance-based pressure sensors, solid-state pressure sensors, manometers, barometers, resistive pressure sensors, etc.

(55) In addition, controller **300** may be connected to the valve **154** positioned along collection pipe **150** and the one or more vents **120**, **160** (note: one vent **120** positioned along collection pipe **150** and one vent **160** positioned on the cap **110** are shown in FIG. **12**, but any one or more of the vents **120** and/or vents **160** shown in FIG. **4** may be included and connected to controller **300** in some

embodiments). For instance, the valve **154** and/or the one or more vents **120, 160** may include motorized or actuatable valves that may be transitioned between the open and closed positions (previously described above) via the controller **300** (or a command or signal generated thereby). In addition, controller **300** may be connected to the valve **154** and/or the one or more vents **120, 160** such that controller **300** may determine a position (such as open, closed, or a position between fully open or fully closed) of the valve **154** and/or the one or more vents **120, 160**. For instance, the valve **154** and/or the one or more vents **120, 160** may include or be connected to a driver such as a stepper motor, servo-motor, or other suitable device that may controllably place the valve **154** and/or the one or more vents **120, 160** in a particular position. The controller **300** may determine a position of the valve **154** and/or the one or more vents **120, 160** via the previous actuation(s) of the drivers (e.g., stepper motor, servo-motor, etc.). In some embodiments, one or more additional sensors or devices, such as position sensors, proximity sensors, pressure sensors (e.g., differential pressure sensors), optical sensors, etc. may be utilized by controller **300** to determine a position of the valve **154** and/or the one or more vents **120, 160** during operations.

(56) Further, controller **300** may be connected to a sensor **324** that is configured to detect whether a conduit (such as conduit **202** shown in FIGS. **7, 8**, and **11C**) is connected to the coupling **152** of collection pipe **150**. The sensor **324** may include a proximity sensor, optical sensor, magnetic sensor, switch, or any other suitable device that may be configured to detect when a conduit (or coupling connected thereto) is engaged with the coupling **152** such that exhaust flowing out of the collection pipe **150** may be directed to another location or system (such as exhaust cleaning assembly **200** shown in FIGS. **7, 8**, and **11A-11C**).

(57) During operations, controller **300** may selectively actuate the valve **154** and/or the vent(s) **120, 160** between the open and closed positions so as to route the exhaust emitted from the one or more exhaust pipes **52** either via the first flow path(s) **122** or the second flow path **124**. For instance, when marine vessel **10** is at berth (e.g., such as shown in FIGS. **9B** and **9C** previously described above), and a conduit (e.g., conduit **202**) is engaged with the coupling **152**, the controller **300** may open the valve **154** and close the vent(s) **120, 160** so that exhaust emitted from the one or more exhaust pipes **52** may be directed along the second flow path **124** and out of the coupling **152** and into an exhaust cleaning assembly (such as exhaust cleaning assembly **200**) as previously described. In addition, when an exhaust cleaning assembly is not connected to the collection pipe **150** (such as via a conduit connected to coupling **152**), such as when the marine vessel **10** is not at berth, the controller **300** may close the valve **154** and open one or more of the vent(s) **120, 160** to allow exhaust emitted from the one or more exhaust pipes **52** to be vented to the surrounding atmosphere **123**. Further, in some embodiments, regardless as to whether the marine vessel **10** is at berth or not at berth (and thus regardless as to whether an exhaust cleaning assembly is connected to the collection pipe **150** via coupling **152**) the controller **300** may actuate one or more of the vents **120, 160** to open (and thereby vent exhaust to the surrounding atmosphere **123**) in response to a determination (e.g., by the controller **300**) that the pressure within the enclosure **112** and/or within the collection pipe **150** is above a threshold. The controller **300** may determine the pressure within the enclosure **112** and within the collection pipe **150** based on outputs received from the sensors **320, 322**. In some embodiments, one or more temperature sensors (e.g., a thermocouple, thermistor, resistance temperature detector (RTD), semiconductor circuit, etc.) may be connected to the exhaust handling system **100** (such as to enclosure **112**, collection pipe **150**, etc.), and the controller **300** (alternatively or additionally) may actuate the one or more vents **120** to open (and thereby vent exhaust to the atmosphere **123**) in response to a determination (such as by the controller **300**) that the temperature within the enclosure **112** and/or the collection pipe **150** is above a threshold.

(58) As shown in FIGS. **10** and **12**, in some embodiments, the exhaust pipes **52** (or some of the exhaust pipes **52**) may be connected to separate collection pipes **159** that are further connected to separate vents **120, 160**, valves **154**, and couplings **152** as previously described. In these embodiments, the controller **300** may be connected to one or more pressure and/or temperature

sensors (such as pressure sensors **320, 322**) that are configured to detect a pressure and/or temperature within the separate collection pipes **159**. In addition, the controller **300** may be connected to the couplings **152** and valves **154** of the separate collection pipes **159**. During operations, the controller **300** may direct exhaust through either the first flow path **122** or the second flow path **124** within each of the separate exhaust pipes **159** via actuation of the separate valves **154** and vent(s) **120, 160** in a similar manner to that previously described for the common collection pipet **150** shown in FIG. **12**.

(59) In some embodiments the vent(s) **120, 160** may be pressure-actuated as previously described. Thus, during operations, one or more of the vents **120, 160** may open or close based on a pressure within the enclosure **112** or collection pipe **150** (or the separate collection pipes **159**) regardless of a position of the valve **154** and/or connection status of the coupling **152** (such as whether the coupling **152** is connected to or not connected to a conduit of an exhaust cleaning assembly such as the conduit **202** and exhaust cleaning assembly **200** shown in FIGS. **7** and **8**). Thus, in some embodiments, the controller **300** may not be connected to the one or more vent(s) **120, 160** of the exhaust handling system **100**.

(60) Reference is now made to FIGS. **13** and **14**, in which methods **400, 450** for flowing the exhaust of a marine vessel through an exhaust handling system (such as the exhaust handling system **100**) are shown according to some embodiments. The method **400** may include a method for flowing the exhaust emitted from the marine vessel (such as marine vessel **10**) along a first flow path to the atmosphere or a second flow path to an exhaust cleaning assembly (such as exhaust cleaning assembly **200** shown in FIGS. **7** and **8**), and the method **450** may include a method for flowing exhaust to the atmosphere based on a pressure within the exhaust handling system (such as exhaust handling system **100**). In some embodiments, the methods **400, 450** may be performed (wholly or partially) by the processor of a controller (such as processor **302** of controller **300** shown in FIG. **12**). Thus, the methods **400, 450** may be representative of the machine-readable instructions **306** stored on memory **304**, or some of the machine-readable instructions **306** for some embodiments of controller **300** (FIG. **12**). In addition, in describing the features of methods **400, 450**, continuing reference will be made to the features of the embodiments shown in FIGS. **1-12** and previously described. In some embodiments, the methods **400, 450** may be performed in parallel (such as by processor **302**), and/or may be combined or integrated with one another.

(61) The method **400** shown in FIG. **13** may initially include a determination, at block **402**, as to whether a coupling of a collection pipe of an exhaust handling system (such as exhaust handling system **100**) is connected to an exhaust cleaning assembly (such as exhaust cleaning assembly **200**). For instance, as shown in FIG. **12** and previously described, the controller **300** may determine whether a conduit **202** of an exhaust cleaning assembly **200** (FIGS. **7** and **8**) is connected to the coupling **152** of collection pipe **150** via an output from the sensor **324** as previously described.

(62) If it is determined that the coupling of the collection pipe is not connected to an exhaust cleaning assembly (the determination at block **402** is “No” or “N”), the method **400** may proceed to flow exhaust emitted from the marine vessel **10** along a first flow path to the atmosphere (for example, atmosphere **123**) at block **404**. For instance, as shown in FIG. **12** and previously described, in some embodiments, the controller **300** may determine that a conduit **202** of an exhaust cleaning assembly **200** is not connected to coupling **152** of collection pipe **150** via an output from the sensor **324**, and may thus close the valve **154** and open one or more of the vent(s) **120, 160** (or allow the vent(s) **120, 160** to open via pressure actuation as previously described) so as to flow the exhaust emitted from the one or more exhaust pipes **52** into the surrounding atmosphere **123** via the one or more first flow paths **122**.

(63) Conversely, if it is determined that the coupling of the collection pipe is connected to an exhaust cleaning assembly (the determination at block **402** is “Yes” or “Y”), the method **400** may proceed to flow exhaust emitted from the marine vessel **10** along a second flow path to the exhaust cleaning assembly at block **406**. For instance, as shown in FIG. **12** and previously described, in

some embodiments, the controller **300** may determine that a conduit **202** is connected is connected to the coupling **152** via an output from the sensor **324**, and may thus may open the valve **154** and (potentially) close the vent(s) **120**, **160** so as to flow the exhaust emitted from the one or more exhaust pipes **52** along the second flow path **124** to the exhaust cleaning assembly **200**.

(64) The method **450** shown in FIG. **14** may initially include receiving an indication of a pressure within an enclosure and/or a collection pipe of an exhaust handling system (such as exhaust handling system **100**) at block **452**. For instance, as shown in FIG. **12** and previously described, the controller **300** may receive an indication of a pressure within the enclosure **112** and/or the collection pipe **150** via outputs from the pressure sensor **320** and/or the pressure sensor **322**, respectively.

(65) Next, method **450** includes determining whether the pressure (such as the pressure within the enclosure **112** and/or the pressure within the collection pipe **150**) is above a threshold at block **454**. For instance, block **454** may include determining whether a particular one or either of the pressures within the enclosure (e.g., enclosure **112**) of the collection pipe (e.g., collection pipe **150**) is above a threshold. In some embodiments, block **454** may include determining whether the pressure within the enclosure (for example, enclosure **112**) is above a first threshold and/or whether the pressure within the collection pipe (for example, collection pipe **150**) is above a second threshold, wherein the first and second thresholds are different. The threshold (or thresholds) may be determined based upon an operating pressure (or pressure range) of the enclosure (for example, enclosure **112**) and/or the collection pipe (e.g., collection pipe **150**).

(66) If it is determined that a pressure within the enclosure and/or the collection pipe is above a threshold (the determination at block **454** is “Yes” or “Y”), method **450** may proceed to open one or more vents to vent exhaust from the enclosure and/or the collection pipe block **456**. For instance, as shown in FIG. **12** and previously described, if the controller **300** determines that the pressure within the exhaust handling system **100** (including the enclosure **112** and/or the collection pipe **150**) is above a threshold, the controller **300** may open one or more of the vent(s) **120** or the vent(s) **160** so as to flow the exhaust out of the enclosure **112** and/or collection pipe **150** to the surrounding atmosphere **123** in an effort to reduce the pressure below the threshold. Conversely, if it is determined that a pressure within the enclosure and/or the collection pipe is not above a threshold (the determination at block **454** is “No” or “N”), method **450** may repeat block **452** and continue monitoring the pressure within the enclosure and/or the collection pipe.

(67) If the vent(s) (for example, vent(s) **120**, vent(s) **160**) are open via block **456**, method **450** may proceed to once again receive an indication of the pressure within the enclosure and/or the collection pipe at block **457** and then determine whether the pressure within the enclosure and/or the collection pipe **150** is below a threshold at block **458**. The indication of the pressure received at block **457** may be obtained in a similar manner to that described above for block **452**. The threshold at block **458** may be same or different from the threshold in block **454**. In some embodiments, the threshold at block **458** may be lower than the threshold at block **454**. The threshold at block **458** may be determined such that the pressure within the enclosure and/or the collection pipe is within a desired or operating range (potentially including a safety factor). If it is determined that the pressure is below the threshold at block **458** (the determination at block **458** is “Yes” or “Y”), method **450** may proceed to close the vent(s) at block **460**. For instance, as shown in FIG. **12** and previously described, if the controller **300** determines that the pressure within the enclosure **112** and/or collection pipe **150** is below a threshold (or within a desired range) (such as via the sensors **320**, **322**) after the vent(s) **120** and/or the vent(s) **160** have been opened to direct exhaust to the atmosphere **123**, the controller **300** may close the vent(s) **120**, **160**. In some embodiments, closing the vent(s) **120**, **160** may allow the exhaust to flow along the second flow path **124** toward the exhaust cleaning assembly (such as exhaust cleaning assembly **200**) as previously described.

(68) Conversely, if it is determined that the pressure within the enclosure and/or the collection pipe

is not below the threshold at block **458** (the determination at block **458** is “No” or “N”), method **450** may repeat back to block **457** to once again receive an indication of the pressure, and then determine whether the pressure within the enclosure and/or the collection pipe is below the threshold at block **458**. For instance, as shown in FIG. **12** and previously described, if after opening the vent(s) **120, 160**, the controller **300** determines is still above a threshold (and is therefore above a desired range), the controller **300** may maintain the vent(s) **120, 160** in the open position and continue to monitor the pressure within the enclosure **112** and/or the collection pipe **150** via the sensors **320, 322** as previously described.

(69) As explained above and reiterated below, this disclosure includes, without limitation, the following example embodiments. Example Embodiment 1: an exhaust handling system for a marine vessel, the exhaust handling system comprising: a cap connected to a top end portion of an exhaust stack of the marine vessel so as to form an enclosure that at least partially surrounds an outlet of an exhaust pipe extending through the exhaust stack; a collection pipe in fluid communication with the cap such that the collection pipe is configured to receive exhaust from the enclosure; and a coupling connected to the collection pipe that is configured to connect to an exhaust cleaning assembly, the exhaust cleaning system including at least one tank to receive the exhaust, the cap at least partially defines a first flow path for the exhaust to flow from the enclosure to an atmosphere surrounding the cap, and the collection pipe at least partially defines a second flow path for the exhaust to flow from the enclosure to the coupling via the collection pipe.

Example Embodiment 2: the exhaust handling system of any example embodiment, wherein the cap is connected to the exhaust stack such that the cap is supported by the exhaust stack. Example Embodiment 3: the exhaust handling system of any example embodiment, wherein the cap is welded to the exhaust stack. Example Embodiment 4: the exhaust handling system of any example embodiment, further comprising a vent, connected to the cap, wherein the vent at least partially defines the first flow path, and wherein the vent includes at least one valve member that is

configured to actuate between: an open position to emit exhaust to the atmosphere via the first flow path; and a closed position to prevent emission of exhaust to the atmosphere via the first flow path.

Example Embodiment 5: the exhaust handling system of any example embodiment, wherein the vent comprises a pressure-actuated vent. Example Embodiment 6: the exhaust handling system of any example embodiment, wherein the at least one valve member comprises a plurality of louvers that are rotatable to transition the vent between the open position and the closed position. Example Embodiment 7: the exhaust handling system of any example embodiment, wherein the coupling is proximate to a deck of the marine vessel. Example Embodiment 8: the exhaust handling system of any example embodiment, further comprising a cleanout port positioned along the collection pipe that is proximate to the deck of the marine vessel, wherein the cleanout port is configured to provide access into the collection pipe independent of the coupling. Example Embodiment 9: the exhaust handling system of any example embodiment, further comprising a pipe connection positioned within the enclosure that is in fluid communication with the exhaust pipe such that exhaust pipe is configured to output exhaust into the pipe connection. Example Embodiment 10: the exhaust handling system of any example embodiment, wherein the pipe connection is connected to the collection pipe such that exhaust emitted from the exhaust pipe is routed into the collection pipe via the pipe connection. Example Embodiment 11: the exhaust handling system of any example embodiment, wherein the collection pipe comprises flexible ductwork. Example Embodiment 12: an exhaust handling system for a marine vessel, the exhaust handling system comprising: a cap connected to a top end portion of an exhaust stack of the marine vessel so as to form an enclosure that at least partially surrounds an outlet of an exhaust pipe extending through the exhaust stack; a vent in fluid communication with the enclosure, the vent including at least one valve member that is actuatable between a first position to emit exhaust from the enclosure to an atmosphere surrounding the cap via the vent and a second position to prevent an emission of exhaust from the enclosure to the atmosphere via the vent; and a collection pipe in fluid

communication with the enclosure; a coupling connected to the collection pipe and configured to connect to an exhaust cleaning assembly that includes at least one tank to receive the exhaust. Example Embodiment 13: the exhaust handling system of any example embodiment, wherein the at least one valve member comprises a plurality of louvers that are rotatable to transition the vent between the first position and the second position. Example Embodiment 14: the exhaust handling system of any example embodiment, wherein the vent comprises a pressure-actuated vent. Example Embodiment 15: the exhaust handling system of any example embodiment, wherein the pressure-actuated vent is positioned on a top end of the cap. Example Embodiment 16: the exhaust handling system of any example embodiment, wherein the pressure-actuated vent is positioned along the collection pipe. Example Embodiment 17: the exhaust handling system of any example embodiment, wherein the cap is connected to the exhaust stack such that the cap is supported by the exhaust stack. Example Embodiment 18: the exhaust handling system of any example embodiment, wherein the cap is welded to the exhaust stack. Example Embodiment 19: the exhaust handling system of any example embodiment, wherein the coupling is proximate to a deck of the marine vessel. Example Embodiment 20: the exhaust handling system of any example embodiment, wherein the collection pipe comprises a cleanout port that is proximate to the deck of the marine vessel, and wherein the cleanout port is configured to provide access into the collection pipe independent of the coupling. Example Embodiment 21: the exhaust handling system of any example embodiment, further comprising a pipe connection positioned within the enclosure that is in fluid communication with the exhaust pipe and the collection pipe such that the exhaust pipe is configured to output exhaust into collection pipe via the pipe connection. Example Embodiment 22: an exhaust handling system for a marine vessel, the exhaust handling system comprising: a cap connected to a top end portion of an exhaust stack of the marine vessel such that the cap is supported by the exhaust stack and such that the cap forms an enclosure that at least partially surrounds an outlet of an exhaust pipes extending through the exhaust stack; a pressure-actuated vent in fluid communication with the enclosure; and a collection pipe in fluid communication with the enclosure and configured to connect to an exhaust cleaning assembly that includes at least one tank to receive the exhaust. Example Embodiment 23: the exhaust handling system of any example embodiment, wherein the pressure-actuated vent includes at least one valve member that is actuatable to selectively emit exhaust from the enclosure to an atmosphere surrounding the cap. Example Embodiment 24: the exhaust handling system of any example embodiment, wherein the pressure-actuated vent is positioned on a top end of the cap. Example Embodiment 25: the exhaust handling system of any example embodiment, wherein the pressure-actuated vent is positioned along the collection pipe. Example Embodiment 26: the exhaust handling system of any example embodiment, wherein the cap is welded to the exhaust stack. Example Embodiment 27: the exhaust handling system of any example embodiment, further comprising a coupling connected to the collection pipe that is configured to connect to a conduit of the exhaust cleaning assembly, wherein the coupling is accessible from a deck of the marine vessel. Example Embodiment 28: the exhaust handling system of any example embodiment, wherein the collection pipe comprises a cleanout port that is accessible from the deck of the marine vessel, and wherein the cleanout port is configured to provide access into the collection pipe independent of the coupling. Example Embodiment 29: the exhaust handling system of any example embodiment, further comprising a pipe connection positioned within the enclosure that is in fluid communication with the exhaust pipe such that the exhaust pipe is configured to output exhaust to the collection pipe via the pipe connection. Example Embodiment 30: a method comprising: (a) positioning a marine vessel in a berthing location, the marine vessel including a deck, an exhaust stack having an top end portion positioned above from the deck, an exhaust pipe extending through the exhaust stack to an outlet that is positioned above the top end portion of the exhaust stack, a cap connected to the exhaust stack so as to form an enclosure that at least partially surrounds the outlet of the exhaust pipe, and an collection pipe in fluid communication with the enclosure; (b) connecting an exhaust cleaning

assembly to the collection pipe after (a), the exhaust cleaning assembly including at least one tank to receive the exhaust; and (c) operating the marine vessel to flow an exhaust emitted from the outlet of the exhaust pipe to the exhaust cleaning assembly after (b) via the collection pipe.

Example Embodiment 31: the method of any example embodiment, further comprising: (d) emitting the exhaust out of the enclosure to an atmosphere surrounding the cap during (a).

Example Embodiment 32: the method of any example embodiment, wherein (d) further comprises flowing the exhaust out of a vent in fluid communication with the enclosure.

Example Embodiment 33: the method of any example embodiment, wherein (d) further comprises flowing the exhaust out of a pressure-actuated vent in fluid communication with the enclosure.

Example Embodiment 34: the method of any example embodiment, further comprising: (e) disconnecting the exhaust cleaning assembly from the collection pipe; (f) moving the marine vessel away from the berthing location after (e); and (g) flowing the exhaust out of the enclosure to the atmosphere during (f).

Example Embodiment 35: the method of any example embodiment, wherein (b) comprises connecting a conduit connected to the exhaust cleaning assembly to a coupling that is connected to the collection pipe and that is proximate to the deck of the marine vessel.

Example Embodiment 36: the method of any example embodiment, further comprising preventing the exhaust from flowing from the enclosure to an atmosphere surrounding the cap during (c).

Example Embodiment 37: the method of any example embodiment, wherein the exhaust cleaning assembly is positioned on a barge that is further positioned at the berthing location.

Example Embodiment 38: a method comprising: (a) positioning a marine vessel in a berthing location, the marine vessel including an exhaust stack, an exhaust pipe extending through the exhaust stack to an outlet that is positioned above a top end portion of the exhaust stack, a cap that is connected to the exhaust stack to form an enclosure that surrounds the outlet of the exhaust pipe; (b) flowing an exhaust from the enclosure to an atmosphere surrounding the cap via a first flow path during (a), the first flow path at least partially defined by the cap; (c) connecting an exhaust cleaning assembly positioned at the berthing location to the cap after (a), the exhaust cleaning system including at least one tank to receive the exhaust; and (d) flowing the exhaust from the enclosure to the exhaust cleaning assembly via a second flow path after (b), the second flow path at least partially defined by the cap.

Example Embodiment 39: the method of any example embodiment, wherein (b) further comprises flowing the exhaust out of a vent connected to the enclosure.

Example Embodiment 40: the method of any example embodiment, wherein (b) further comprises rotating at least one louver of the vent to open the vent.

Example Embodiment 41: the method of any example embodiment, further comprising: (e) disconnecting the exhaust cleaning assembly from the cap; (f) moving the marine vessel away from the berthing location after (e); and (g) flowing the exhaust out of the enclosure to the atmosphere via the first flow path during (f).

Example Embodiment 42: the method of any example embodiment, wherein (c) comprises connecting a conduit connected to the exhaust cleaning assembly to a coupling that is proximate to a deck of the marine vessel and that is in fluid communication with the enclosure.

Example Embodiment 43: the method of any example embodiment, further comprising preventing the exhaust from flowing from the enclosure to the atmosphere during (d).

Example Embodiment 44: the method of any example embodiment, wherein the exhaust cleaning assembly is positioned on a barge that is positioned at the berthing location.

(70) As described above, the embodiments disclosed herein include exhaust handling systems for a marine vessel that are configured to allow selective collection of exhaust output from the exhaust-emitting systems of the marine vessel while at a berthing location. In some embodiments, the collected exhaust may be routed to an exhaust cleaning assembly that may store and/or process the exhaust to prevent harmful chemicals or pollutants therein from being emitted to the atmosphere. In some embodiments, the exhaust handling systems of the embodiments disclosed herein may include a cap that is connected to an upper end portion of an exhaust stack of the marine vessel so as to form an enclosure around one or more (e.g., a plurality of) exhaust pipes that emit exhaust

from the exhaust-emitting systems of the marine vessel. The collected exhaust may be emitted directly to the atmosphere when the marine vessel is not at a berthing location (or during a situation that requires venting of exhaust to the atmosphere), or may be selectively routed to an exhaust cleaning assembly to avoid such atmospheric venting when the marine vessel is berthed. Accordingly, through use of the embodiments disclosed herein, the exhaust-emitting systems of a marine vessel may continue to operate when the marine vessel is berthed while preventing (or restricting) the emission of exhaust (or at least the harmful and/or polluting components thereof) to the atmosphere.

(71) The discussion above is directed to various exemplary embodiments. However, one of ordinary skill in the art will understand that the examples disclosed herein have broad application, and that the discussion of any embodiment is meant only to be exemplary of that embodiment, and not intended to suggest that the scope of the disclosure, including the claims, is limited to that embodiment.

(72) The drawing figures are not necessarily to scale. Certain features and components herein may be shown exaggerated in scale or in somewhat schematic form and some details of conventional elements may not be shown in interest of clarity and conciseness.

(73) In the discussion herein and in the claims, the terms “including” and “comprising” are used in an open-ended fashion, and thus should be interpreted to mean “including, but not limited to” Also, the terms “couple,” “couples,” “connect,” or “connection,” are intended to mean either an indirect or direct connection. Thus, if a first device couples or connects to a second device, that connection may be through a direct connection of the two devices, or through an indirect connection that is established via other devices, components, nodes, and connections. In addition, when used herein (including in the claims), the words “about,” “generally,” “substantially,” “approximately,” and the like, when used in reference to a stated value mean within a range of plus or minus 10% of the stated value. Further, as used herein, the terms “axial” and “axially” generally mean along or parallel to a given axis (e.g., central axis of a body or a port), while the terms “radial” and “radially” generally mean perpendicular to the given axis. For instance, an axial distance refers to a distance measured along or parallel to the axis, and a radial distance means a distance measured perpendicular to the axis.

(74) This application is a continuation of U.S. Non-Provisional application Ser. No. 18/369,485, filed Sep. 18, 2023, titled “Exhaust Handling Systems for Marine Vessels and Related Methods,” which claims priority to and the benefit of U.S. Provisional Application No. 63/485,886, filed Feb. 18, 2023, titled “Exhaust Handling Systems for Marine Vessels and Related Methods,” and U.S. Provisional Application Ser. No. 63/488,574, filed Mar. 6, 2023, titled “Exhaust Handling Systems for Marine Vessels and Related Methods,” the contents of which are incorporated herein by reference in their entirety.

(75) While exemplary embodiments have been shown and described, modifications thereof can be made by one skilled in the art without departing from the scope or teachings herein. The embodiments described herein are exemplary only and are not limiting. Many variations and modifications of the systems, apparatus, and processes described herein are possible and are within the scope of the disclosure. Accordingly, the scope of protection is not limited to the embodiments described herein, but is only limited by the claims that follow, the scope of which shall include all equivalents of the subject matter of the claims. Unless expressly stated otherwise, the steps in a method claim may be performed in any order. The recitation of identifiers such as (a), (b), (c) or (1), (2), (3) before steps in a method claim are not intended to and do not specify a particular order to the steps, but rather are used to simplify subsequent reference to such steps.

Claims

1. An exhaust handling system for a marine vessel, the exhaust handling system comprising: a cap connected to a top end portion of an exhaust stack of the marine vessel so as to define an enclosure positioned to at least partially surround an outlet of an exhaust pipe extending through the exhaust stack of the marine vessel; a collection pipe in fluid communication with the cap and positioned to receive exhaust from the enclosure so that the cap at least partially defines a first flow path for the exhaust to flow from the enclosure to an atmosphere surrounding the cap; a coupling connected to the collection pipe so that the collection pipe at least partially defines a second flow path for the exhaust to flow from the enclosure to the coupling via the collection pipe; and an exhaust cleaning assembly connected to the coupling and including at least one tank to receive the exhaust.
2. The exhaust handling system of claim 1, wherein the cap is connected to the exhaust stack such that the cap is supported by the exhaust stack.
3. The exhaust handling system of claim 1, further comprising a vent, connected to the cap, wherein the vent also at least partially defines the first flow path, and wherein the vent includes at least one valve member positioned to operate between: an open position to emit exhaust to the atmosphere via the first flow path, and a closed position to prevent emission of exhaust to the atmosphere via the first flow path.
4. The exhaust handling system of claim 3, wherein the vent comprises a pressure-actuated vent.
5. The exhaust handling system of claim 4, wherein the at least one valve member comprises a plurality of louvers rotatable to transition the vent between the open position and the closed position.
6. The exhaust handling system of claim 1, wherein the coupling is positioned proximate to a deck of the marine vessel.
7. The exhaust handling system of claim 6, further comprising a cleanout port positioned along the collection pipe proximate to the deck of the marine vessel, and wherein the cleanout port provides access into the collection pipe independent of the coupling.
8. The exhaust handling system of claim 1, further comprising a pipe connection positioned within the enclosure and in fluid communication with the exhaust pipe such that exhaust pipe is outputs exhaust into the pipe connection, and wherein the pipe connection is connected to the collection pipe such that exhaust emitted from the exhaust pipe is routed into the collection pipe via the pipe connection.
9. An exhaust handling system for a marine vessel, the exhaust handling system comprising: a cap connected to a top end portion of an exhaust stack of the marine vessel such that the cap is supported by the exhaust stack and such that the cap defines an enclosure positioned to at least partially surround an outlet of an exhaust pipe extending through the exhaust stack of the marine vessel; a pressure-actuated vent in fluid communication with the enclosure; and a collection pipe in fluid communication with the enclosure to receive exhaust therefrom and connected to an exhaust cleaning assembly, the exhaust cleaning assembly including at least one tank to receive the exhaust.
10. The exhaust handling system of claim 9, wherein the pressure-actuated vent includes at least one valve member actuatable to selectively emit exhaust from the enclosure to an atmosphere surrounding the cap during operation.
11. The exhaust handling system of claim 9, wherein the pressure-actuated vent is positioned on a top end of the cap, and wherein the pressure-actuated vent is positioned along the collection pipe.
12. The exhaust handling system of claim 9, further comprising a coupling connected to the collection pipe and positioned to connect to a conduit of the exhaust cleaning assembly, wherein the coupling is accessible from a deck of the marine vessel, wherein the collection pipe comprises a cleanout port that is accessible from the deck of the marine vessel, and wherein the cleanout port is positioned to provide access into the collection pipe independent of the coupling.
13. The exhaust handling system of claim 9, further comprising a pipe connection positioned within the enclosure and in fluid communication with the exhaust pipe such that the exhaust pipe outputs

exhaust to the collection pipe via the pipe connection during operation.

14. A method comprising: (a) positioning a marine vessel in a berthing location, the marine vessel including a deck, an exhaust stack having a top end portion positioned above the deck, an exhaust pipe extending through the exhaust stack to an outlet and positioned above the top end portion of the exhaust stack, a cap connected to the exhaust stack so as to define an enclosure positioned to at least partially surround the outlet of the exhaust pipe, and a collection pipe in fluid communication with the enclosure to receive exhaust therefrom; (b) connecting one or more tanks to the collection pipe after step (a), thereby to receive the exhaust; and (c) operating the marine vessel to flow an exhaust emitted from the outlet of the exhaust pipe to the one or more tanks after step (b) and via the collection pipe.

15. The method of claim 14, further comprising: (d) emitting the exhaust out of the enclosure through a vent to atmosphere surrounding the cap during step (a) when the marine vessel is in operation.

16. The method of claim 15, wherein step (d) comprises flowing the exhaust out of a pressure-actuated vent in fluid communication with the enclosure.

17. The method of claim 14, further comprising: (e) disconnecting the exhaust cleaning assembly from the collection pipe; (f) moving the marine vessel away from the berthing location after step (e); and (g) flowing the exhaust out of the enclosure to atmosphere during step (f).

18. The method of claim 14, wherein step (b) comprises connecting a conduit connected to the one or more tanks to a coupling, the coupling connected to the collection pipe and positioned proximate to the deck of the marine vessel.

19. The method of claim 14, further comprising preventing the exhaust from flowing from the enclosure to atmosphere surrounding the cap during step (c).

20. A method comprising: (a) positioning a marine vessel in a berthing location, the marine vessel including an exhaust stack, an exhaust pipe extending through the exhaust stack to an outlet positioned at an elevation above a top end portion of the exhaust stack, and a cap connected to the exhaust stack to define an enclosure positioned to surround the outlet of the exhaust pipe; (b) flowing an exhaust from the enclosure to atmosphere surrounding the cap via a first flow path during step (a), the first flow path at least partially defined by the cap; (c) connecting one or more tanks to receive the exhaust; and (d) flowing the exhaust from the enclosure to the one or more tanks via a second flow path after step (b), the second flow path at least partially defined by the cap.

21. The method of claim 20, wherein step (b) further comprises flowing the exhaust out of a vent connected to the enclosure, and wherein the vent includes at least one louver, and wherein step (b) further comprises rotating the at least one louver of the vent to open the vent.

22. The method of claim 20, further comprising: (e) disconnecting the exhaust cleaning assembly from the cap; (f) moving the marine vessel away from the berthing location after step (e); and (g) flowing the exhaust out of the enclosure to the atmosphere via the first flow path during step (f).

23. The method of claim 20, wherein step (c) comprises connecting a conduit connected to the one or more tanks to a coupling positioned proximate to a deck of the marine vessel and in fluid communication with the enclosure, and the method further comprising preventing the exhaust from flowing from the enclosure to the atmosphere during step (d).

24. The method of claim 20, wherein the one or more tanks is positioned on a barge positioned adjacent the berthing location.
